

## Unit 10, Lesson 9: Is it Linear?

### Benchmarks

**MA.8.F.1.1** Given a set of ordered pairs, a table, a graph, or mapping diagram, determine whether the relationship is a function. Identify the domain and range of the relation.

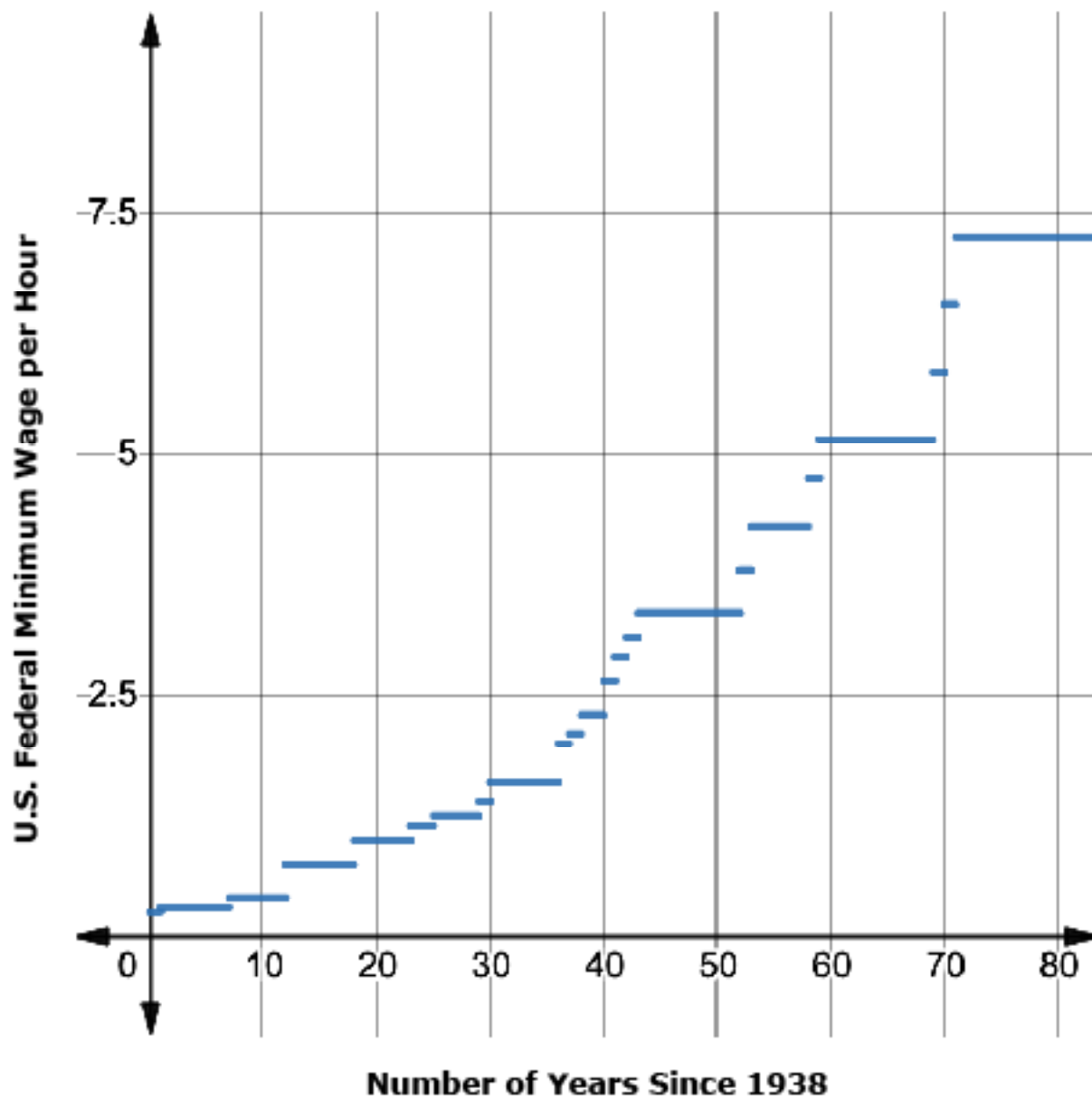
**MA.8.F.1.2** Given a function defined by a graph or an equation, determine whether the function is a linear function. Given an input-output table, determine whether it could represent a linear function.

### Learning Targets

- ☐ I can determine whether a function is a linear function given an equation or graph.
- ☐ I can determine whether a relation is a function.

**10.9.1 Warm-Up: Notice and Wonder**

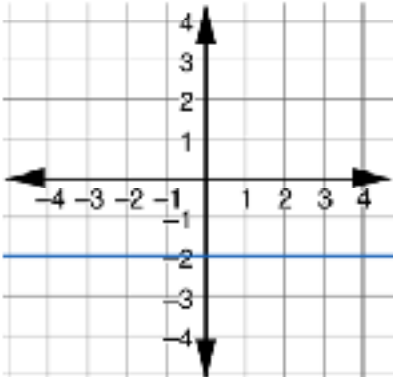
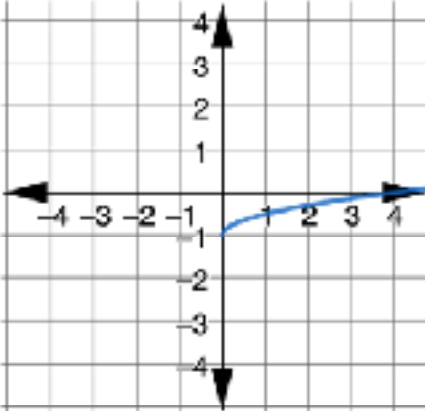
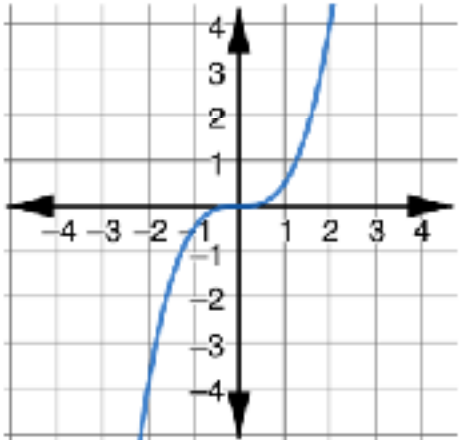
1. The graph shows how the value of the U.S. federal hourly minimum wage changed from 1938–2020.

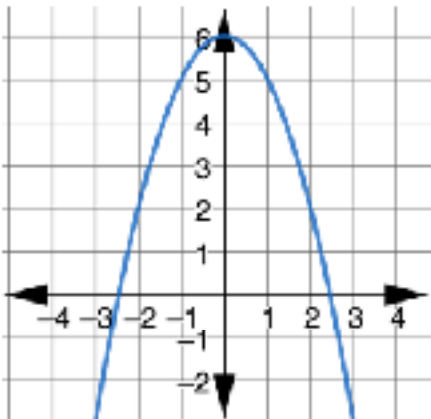
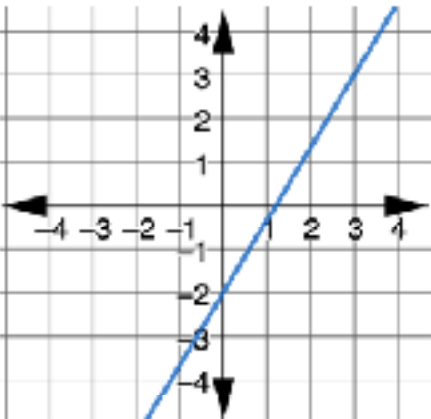
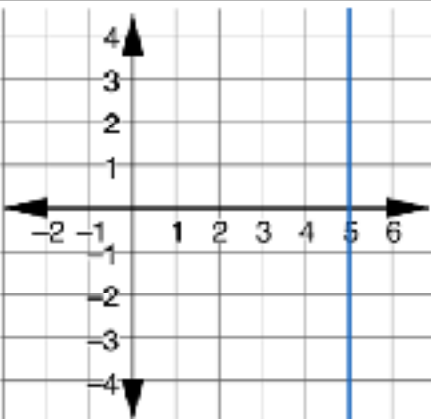
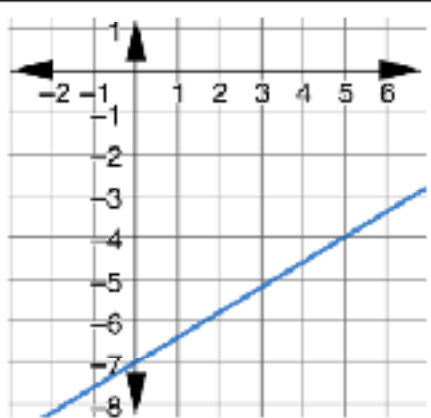


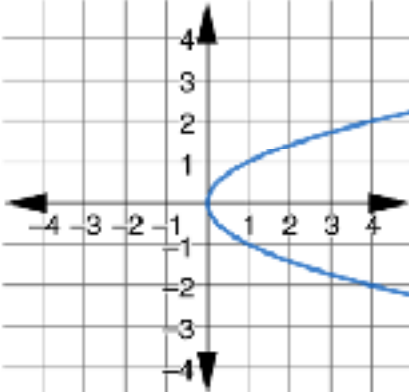
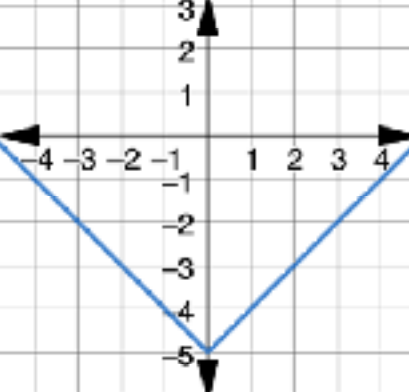
- a. What do you notice?
- b. What do you wonder?

10.9.2 Collaboration: Revising Conjectures about Non-Linear Equations

1. Work with your partner to answer the questions for the description of each equation and its graph.

| Equation |                               | Graph                                                                                | Description                                                                                                                                                                                                                                                         |
|----------|-------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A        | $y = -2$                      |     | <p>Is it linear?</p> <ul style="list-style-type: none"><li><input type="radio"/> Yes</li><li><input type="radio"/> No</li></ul> <p>Is it a function?</p> <ul style="list-style-type: none"><li><input type="radio"/> Yes</li><li><input type="radio"/> No</li></ul> |
| B        | $y = \frac{1}{2}\sqrt{x} - 1$ |   | <p>Is it linear?</p> <ul style="list-style-type: none"><li><input type="radio"/> Yes</li><li><input type="radio"/> No</li></ul> <p>Is it a function?</p> <ul style="list-style-type: none"><li><input type="radio"/> Yes</li><li><input type="radio"/> No</li></ul> |
| C        | $\frac{x^3}{2} = y$           |  | <p>Is it linear?</p> <ul style="list-style-type: none"><li><input type="radio"/> Yes</li><li><input type="radio"/> No</li></ul> <p>Is it a function?</p> <ul style="list-style-type: none"><li><input type="radio"/> Yes</li><li><input type="radio"/> No</li></ul> |

|   |                        |                                                                                      |                                                                                                                                                                                                                                                                     |
|---|------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D | $y = -x^2 + 6$         |    | <p>Is it linear?</p> <ul style="list-style-type: none"><li><input type="radio"/> Yes</li><li><input type="radio"/> No</li></ul> <p>Is it a function?</p> <ul style="list-style-type: none"><li><input type="radio"/> Yes</li><li><input type="radio"/> No</li></ul> |
| E | $5x = 6 + 3y$          |   | <p>Is it linear?</p> <ul style="list-style-type: none"><li><input type="radio"/> Yes</li><li><input type="radio"/> No</li></ul> <p>Is it a function?</p> <ul style="list-style-type: none"><li><input type="radio"/> Yes</li><li><input type="radio"/> No</li></ul> |
| F | $x = 5$                |  | <p>Is it linear?</p> <ul style="list-style-type: none"><li><input type="radio"/> Yes</li><li><input type="radio"/> No</li></ul> <p>Is it a function?</p> <ul style="list-style-type: none"><li><input type="radio"/> Yes</li><li><input type="radio"/> No</li></ul> |
| G | $y = \frac{3}{5}x - 7$ |  | <p>Is it linear?</p> <ul style="list-style-type: none"><li><input type="radio"/> Yes</li><li><input type="radio"/> No</li></ul> <p>Is it a function?</p> <ul style="list-style-type: none"><li><input type="radio"/> Yes</li><li><input type="radio"/> No</li></ul> |

|   |                       |                                                                                      |                                                                                                                                                                                                                                                                           |
|---|-----------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| H | $x = y^2$             |    | <p>Is it linear?</p> <ul style="list-style-type: none"> <li><input type="radio"/> Yes</li> <li><input type="radio"/> No</li> </ul> <p>Is it a function?</p> <ul style="list-style-type: none"> <li><input type="radio"/> Yes</li> <li><input type="radio"/> No</li> </ul> |
| I | $y = 3^x - 2$         |   | <p>Is it linear?</p> <ul style="list-style-type: none"> <li><input type="radio"/> Yes</li> <li><input type="radio"/> No</li> </ul> <p>Is it a function?</p> <ul style="list-style-type: none"> <li><input type="radio"/> Yes</li> <li><input type="radio"/> No</li> </ul> |
| J | $y = \frac{2}{x} + 1$ |  | <p>Is it linear?</p> <ul style="list-style-type: none"> <li><input type="radio"/> Yes</li> <li><input type="radio"/> No</li> </ul> <p>Is it a function?</p> <ul style="list-style-type: none"> <li><input type="radio"/> Yes</li> <li><input type="radio"/> No</li> </ul> |
| K | $y =  x  - 5$         |  | <p>Is it linear?</p> <ul style="list-style-type: none"> <li><input type="radio"/> Yes</li> <li><input type="radio"/> No</li> </ul> <p>Is it a function?</p> <ul style="list-style-type: none"> <li><input type="radio"/> Yes</li> <li><input type="radio"/> No</li> </ul> |

Work with your partner to answer the following questions based on your observations of the relations in the table.

2. Which equations are **not** functions? Write the equations.
  
  
  
  
  
  
  
  
  
  
3. Explain how to use the graph of an equation to determine if the equation represents a function.
  
  
  
  
  
  
  
  
  
  
4. What do you notice about the variables  $x$  and  $y$  in the equations that are linear?
  
  
  
  
  
  
  
  
  
  
5. Write at least one phrase that could be used to complete the statement below.

*An equation is **not** linear if either variable is ...*

6. The equations of several relations are organized in the table. Complete the statement to describe the graphs of these equations.

| Group A                                                                                                                                                                                                                                                                                                                                               | Group B                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $y = -2$<br>$y = 5$<br>$y = \frac{4}{5}$<br>$y = -21$                                                                                                                                                                                                                                                                                                 | $x = -2$<br>$x = 5$<br>$x = \frac{4}{5}$<br>$x = -21$                                                                                                                                                                                                                                                                                                 |
| <p>The graphs of these equations are</p> <div> <input type="checkbox"/> vertical<br/> <input type="checkbox"/> horizontal         </div> <p>lines</p> <p>because the</p> <div> <input type="checkbox"/> <math>x</math>-values<br/> <input type="checkbox"/> <math>y</math>-values         </div> <p>at every point on the graph will be the same.</p> | <p>The graphs of these equations are</p> <div> <input type="checkbox"/> vertical<br/> <input type="checkbox"/> horizontal         </div> <p>lines</p> <p>because the</p> <div> <input type="checkbox"/> <math>x</math>-values<br/> <input type="checkbox"/> <math>y</math>-values         </div> <p>at every point on the graph will be the same.</p> |

7. Complete the statement.

The equations in Group B

- ☐ are  
☐ are not

functions

because

- ☐ each  $x$ -value  
☐ at least one  $x$ -value

corresponds to

- ☐ exactly one  $y$ -value  
☐ more than one  $y$ -value

on the graph.

### 10.9.3 Bring It Together: Revising Conjectures About Non-Linear Equations

1. Ana made several statements after completing the collaboration with her partner. Some of her statements are not true. Cross out any false statements.

| Ana's Statements                                                                            |
|---------------------------------------------------------------------------------------------|
| "An equation is not linear if both variables are raised to a power of 1."                   |
| "An equation is not linear if either variable is raised to a power other than 1."           |
| "An equation is not linear if a constant term is set equal to $x$ ."                        |
| "An equation is not linear if a variable is an exponent."                                   |
| "An equation is not linear if a variable is in the denominator."                            |
| "An equation is not linear if it includes the absolute value or square root of a variable." |

2. Complete the statements.

- a. In a two-variable equation, the equation represents a linear function if both variables are raised to the \_\_\_\_\_ power, and neither variable is in the 

☐ denominator  
☐ numerator

 or an exponent.
- b. A constant value set equal to 

☐  $x$   
☐  $y$

 can also represent a linear function.
- c. An equation represents a linear function if it can be written in the form  $y = \underline{\hspace{2cm}}$ , where  $m$  is the slope and  $b$  is the  $y$ -intercept.



**10.9.4 Your Turn!**

1. Complete the table.

| Equation               | Linear Function          | Not a Linear Function    |
|------------------------|--------------------------|--------------------------|
| $y = 3x^2 + 5x$        | <input type="checkbox"/> | <input type="checkbox"/> |
| $y = 4 + \frac{1}{6}x$ | <input type="checkbox"/> | <input type="checkbox"/> |
| $y = 3,278$            | <input type="checkbox"/> | <input type="checkbox"/> |
| $y = \sqrt{2x + 9}$    | <input type="checkbox"/> | <input type="checkbox"/> |
| $y = x^4 - 7$          | <input type="checkbox"/> | <input type="checkbox"/> |
| $x = -24$              | <input type="checkbox"/> | <input type="checkbox"/> |
| $-4x = 9y + 13$        | <input type="checkbox"/> | <input type="checkbox"/> |
| $y = 5^{x+1}$          | <input type="checkbox"/> | <input type="checkbox"/> |

2. Choose one equation from the table and explain why it is not a linear function.

3. Choose one equation from the table and explain why it is a linear function.

**Complete each statement as you reflect on today's learning.**

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